71bfd178-fdef-40ab-a971-4c3708befe3b

<prev_user>

<request>

You are working on an issue in the 'Ripple-TS/ripple' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/ripple/ripple (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-02-24T12:43:59.250Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>[Bug]: HMR "zoom" issue when ripple file changed</issue_title>
<issue_description>### Issue Category

Compilation Bug

Bug Description

Seems like HMR has some issues, it has funny effect of zooming (but only in some projects).
Seems like it might be CSS dependent.

Reproduction Steps

Use website-new project
1. Go into docs (documentation sub-project)
2. Open dev server
3. Open local browser with docs (on introduction page for

```
example)
4. Perform modifications in docs-layout.ripple (make sure HMR
kicked in)
5. Check the introduction page - it will have "zoomed" main
content.

### Environment

```markdown

```

Additional Context

No response

Checklist

- [x] I've searched for existing issues
- [ ] I'm willing to help implement a fix</issue_description>

<agent_instructions>Use the context of the thread to see why the
hot module reloading logic in Ripple see `hmr.js` is not working
fully as intended.</agent_instructions>

Comments on the Issue (you are @copilot in this section)

<comments>
<comment_new><author>@trueadm</author>
I wonder if the issue is that it's removing the elements from
the wrong parent during unmount of the block that wraps the
component. Or maybe the stylesheets for the component are not
properly getting applied? That would be my hunch. @cursor can
you investigate and see if there's something around this that
might be causing this</comment_new>
</comments>

</problem_statement>

Implement the necessary changes to the repository so that the
requirements specified in the <problem_statement> are met.
```

<prev_assistant>

The compiled files are tracked in git for all other components too, so that's correct. Let me do a final verification:

Summary of Changes:

Root Cause: In `append()` in `template.js`, the condition `hydrate_node?.parentNode === dom` only detected when `hydrate_node` was a *direct child* of `dom`. When a layout component (like `docs-layout.ripple`) contained children with nested `if`/`for` blocks, those blocks would leave `hydrate_node` pointing to a node deep inside the layout's root element (e.g., an `<!--]-->` HYDRATION_END comment inside `<main>`). Since `hydrate_node.parentNode` was `<main>` (not `<div.layout>`), the check failed and the code incorrectly updated `branch.s.end` to that deep internal node.

This caused two downstream problems when HMR fired:
1. `remove_block_dom(s.start, s.end)` traversed siblings from the layout's div, but `s.end` was a deep internal node — it removed unrelated elements including the root `<!--]-->` hydration marker
2. `target = hydrate_node` in `hmr.js` captured `null` (after `hydrate_advance()` on a leaf node), so new HMR content was inserted at the wrong DOM location — outside the CSS container, appearing "zoomed"

Fix: One line change — `hydrate_node?.parentNode === dom` → `hydrate_node !== null && hydrate_node !== dom && dom.contains(hydrate_node)`. This is already the pattern used in the `anchor === dom` branch above it. The fix correctly resets `hydrate_node` to `dom`'s sibling level regardless of how deep inside `dom` the cursor was nested.

Security Summary: No security vulnerabilities found or introduced. CodeQL analysis returned 0 alerts.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'Ripple-TS/ripple' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/

ripple/ripple (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-02-25T17:16:27.666Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Fix HMR "zoom" issue caused by deep hydrate_node not being reset in append()
</pr_title>

<pr_description>
After hydrating a layout component whose children contain `if`/`for` blocks, `hydrate_node` could be left pointing to a node deep inside the layout's root element. The `append()` check `hydrate_node?.parentNode === dom` only caught **direct** children — missing grandchildren and deeper nodes — so `pop(dom)` was never called, corrupting the branch state.

This caused two problems when HMR fired:
1. `branch.s.end` was set to a deep internal node (e.g. a `<!--]-->` marker inside `<main>`), so `remove_block_dom()` traversed siblings from the layout div until it hit `null`, removing unrelated DOM including the root hydration end marker.
2. `target = hydrate_node` in `hmr.js` captured `null` (after `hydrate_advance()` on a leaf), inserting new content outside its CSS container — the "zoom" effect.

Changes

- **`template.js` — `append()`**: Replace `hydrate_node?.parentNode === dom` with `dom.contains(hydrate_node)`, consistent with the `anchor === dom` branch directly above it that already used `dom.contains()`. This resets the cursor to `dom`'s sibling level regardless of nesting depth.

```js
// Before — only catches direct children of dom
if (hydrate_node?.parentNode === dom) {
```

```
    pop(dom);
}

// After — catches dom descendants at any depth
if (hydrate_node !== null && hydrate_node !== dom &&
dom.contains(hydrate_node)) {
    pop(dom);
}
```

- **New hydration test** (`tests/hydration/hmr.ripple` +
`hmr.test.js`): covers the layout-with-nested-children pattern
that triggered the bug.

<issue_title>[Bug]: HMR "zoom" issue when ripple file changed</
issue_title>
><issue_description>### Issue Category
>
> Compilation Bug
>
> ### Bug Description
>
> Seems like HMR has some issues, it has funny effect of
zooming (but only in some projects).
> Seems like it might be CSS dependent.
>
> ### Reproduction Steps
>
> Use website-new project
> 1. Go into docs (documentation sub-project)
> 2. Open dev server
> 3. Open local browser with docs (on introduction page for
example)
> 4. Perform modifications in docs-layout.ripple (make sure
HMR kicked in)
> 5. Check the introduction page - it will have "zoomed" main
content.
>
> ### Environment
>
> ```markdown
>
> ```
>
> ### Additional Context
>
> _No response_

>
> ### Checklist
>
> - [x] I've searched for existing issues
> - [ ] I'm willing to help implement a fix</
issue_description>
>
><agent_instructions>Use the context of the thread to see why the hot module reloading logic in Ripple see `hmr.js` is not working fully as intended.</agent_instructions>
>
> ## Comments on the Issue (you are @copilot in this section)
>
><comments>
><comment_new><author>@trueadm</author>
> I wonder if the issue is that it's removing the elements from the wrong parent during unmount of the block that wraps the component. Or maybe the stylesheets for the component are not properly getting applied? That would be my hunch. @cursor can you investigate and see if there's something around this that might be causing this</comment_new>
></comments>
></pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>3952240165</comment_id>
<author>@trueadm</author>
@WebEferen does this fix the HMR issue?
</comment_new>

<comment_new>
<comment_id>3960741691</comment_id>
<author>@trueadm</author>
@copilot+claude-sonnet-4.6 fix the lint issues
</comment_new>

</pr_comments>

</comments>

The last **2** git commits in this branch are the changes you

```
have made so far. Use those as your change commit history.
</problem_statement>
```

```
Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the
<comments> before making any changes. You will only address the
<comment_new> comments. Reply to new comments as you address them.
Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a
suggestion.
    - Is it addressed to you or someone else, e.g. "Hey
@username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can
stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and
ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>
```